

# Pawan Eknath Shinde

+91 8999387495 | pawanshinde1130@gmail.com | linkedin.com/in/pawan-shinde-ps12 | github.com/pawanshinde7218

## PROFESSIONAL SUMMARY

Junior Robotics Software Engineer with 6 months of hands-on industry experience developing autonomous mobile robot (AMR) applications using ROS 2, C++, and Python on Ubuntu Linux. Proven expertise across the full AMR software stack — URDF/Xacro robot modeling, Gazebo simulation, sensor fusion (LiDAR, IMU), SLAM-based mapping, AMCL localization, and autonomous navigation with Navigation2 (Nav2). Developed and validated novel variance-based localization logic to solve symmetric-corridor ambiguity — a known production challenge in warehouse robotics. Experienced in ROS 2 node architecture, custom messages/services/actions, TF2, launch file automation, serial communication with embedded hardware, and parameter tuning for path planning and motion planning algorithms. Eager to contribute to ROS 2-based AMR, industrial automation, or robot software development teams.

## TECHNICAL SKILLS

**Languages:** C++, Python

**Robotics Frameworks:** ROS 2 (Humble), Navigation2 (Nav2), ros2\_control (Basic)

**SLAM & Localization:** SLAM Toolbox, Cartographer, AMCL, Extended Kalman Filter (EKF), Sensor Fusion

**Navigation & Planning:** NavFn Planner, DWB Controller, Costmap2D, Path Planning, Motion Planning, Obstacle Avoidance, Behavior Trees (Nav2 BT)

**Robot Modeling & Simulation:** URDF/Xacro, Gazebo (plugins, sensor models), RViz, Differential Drive

**Perception & Sensors:** 2D LiDAR, IMU, Depth Camera, Point Cloud Processing, Robot Perception

**ROS 2 Concepts:** Nodes, Topics, Publishers/Subscribers, Services, Actions, Parameters, Launch Files, Lifecycle Nodes, TF2, Custom Msg/Srv/Actions, DDS Middleware

**Tools & DevOps:** Ubuntu Linux, Git, VS Code, Docker (Basic)

## EXPERIENCE

**Robotics Software Engineering Intern** · Kody Technolab Pvt. Ltd., Ahmedabad, India

Jan 2026 – Jun 2026

- Designed and deployed custom ROS 2 nodes in C++ and Python for a differential drive AMR, implementing robot monitoring, battery management via serial communication (UART) with an STM32-based embedded controller, and autonomous navigation control.
- Engineered custom ROS 2 interfaces (messages, services, and actions) enabling modular communication between navigation, monitoring, and hardware abstraction layers of the robot software architecture.
- Built and simulated a sensor-fused differential drive robot in Gazebo using URDF/Xacro, integrating 2D LiDAR, IMU, and Depth Camera plugins to replicate real-world robot perception pipelines.
- Implemented a complete autonomous navigation pipeline — SLAM Toolbox for occupancy grid mapping, AMCL for Monte Carlo localization, and Navigation2 (Nav2) for path planning, motion planning, and obstacle avoidance — and tuned NavFn Planner, DWB Controller, and Costmap2D parameters to improve navigation reliability.
- Automated full robot bringup (simulation, mapping, localization, and navigation) using modular ROS 2 launch files, reducing manual setup steps and improving team workflow reproducibility.
- Investigated production localization failures caused by symmetric corridor environments and contributed engineering solutions evaluated through structured Gazebo simulation experiments.

## PROJECTS

### Variance-Based Adaptive AMCL Localization for Symmetric Corridor Environments

*Stack: ROS 2, Python, Nav2, AMCL, SLAM Toolbox, Gazebo, RViz, 2D LiDAR*

- Problem:** Standard AMCL particle filters fail in symmetric long-corridor environments due to ambiguous LiDAR scan distributions — a known production failure mode in warehouse AMR deployments.
- Recreated a symmetric corridor environment in Gazebo to reproduce and systematically characterize the localization ambiguity observed in real autonomous mobile robots.
- Developed a variance-based ROS 2 Python node that analyzes 2D LiDAR scan data to classify the robot's environment as symmetric or asymmetric, enabling context-aware localization management.

- Implemented dynamic AMCL parameter tuning via ROS 2 services — selecting particle count, laser model weights, and resampling thresholds based on detected environment symmetry — improving localization robustness without manual intervention.
- Validated the approach by comparing particle filter convergence behavior across symmetric and asymmetric environments using Navigation2, SLAM Toolbox, and Gazebo simulation, confirming improved pose estimation stability.

### Full-Stack Autonomous Mobile Robot: Modeling, Mapping, Localization & Navigation

**Stack:** ROS 2, C++, Python, Nav2, SLAM Toolbox, AMCL, Gazebo, RViz, URDF/Xacro, TF2, LiDAR, IMU, Depth Camera

- Designed a differential drive robot model in URDF/Xacro with accurate inertia parameters and integrated Gazebo plugins for motor simulation and onboard sensor emulation (LiDAR, IMU, Depth Camera).
- Selected SLAM Toolbox over Cartographer for online occupancy grid mapping, prioritizing lifelong mapping support and ROS 2 native compatibility for autonomous environment exploration.
- Resolved a TF2 transform tree inconsistency between the odom and map frames during AMCL initialization by correcting static transform publishers in the launch architecture, enabling stable localization startup.
- Implemented autonomous path planning, motion planning, and obstacle avoidance using Navigation2, configuring Costmap2D layers, the NavFn global planner, and the DWB local controller for reliable goal-reaching in a custom Gazebo environment.
- Developed modular ROS 2 launch files to automate the complete robot pipeline — simulation bringup, SLAM mapping, AMCL localization, and Nav2 navigation — enabling single-command deployment and reproducible testing.

### EDUCATION

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|----------------------------------------------------|----------|
| <b>B.Tech — Automation &amp; Robotics</b>          | Jun 2026 |
| Savitribai Phule Pune University · CGPA: 8.68 / 10 |          |
| <b>HSC (Higher Secondary)</b>                      | Mar 2021 |
| Super Thirty Junior College · 95.00%               |          |
| <b>SSC (Secondary)</b>                             | Mar 2019 |
| Vidya Vikas Madhyamic Vidyalay · 83.60%            |          |

### CERTIFICATIONS

- Robotics Internship Completion Certificate — Kody Technolab Pvt. Ltd.
- Python Programming — Infosys Springboard